

Vol 15 Architectural Scaffold v0.1 — Methodology: BST Investigation + CSE + CI Collaboration

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Filed: 2026-05-23 Saturday ~12:08 EDT (Keeper, scaffold completes 6/6 Wave 2 milestone) Companion to: INDEX.md (chapter outline + reader target)

Anchoring research artifacts (existing coverage by chapter)

Ch 1 AC(0) Bounded-Depth Counting (~50%)

Anchors: T29 closed AC(0) argument + depth ceiling theorems + Casey's Curvature Principle - T421: depth ≤ 1 under Casey strict (proved) - T316: depth $\leq \text{rank} = 2$ (proved) - T29 $P \neq NP$ via AC(0): closed April 23, three independent proof routes (resolution PROVED + all-P CONDITIONAL + curvature route) - Casey's Curvature Principle: "You can't linearize curvature" — $P \neq NP$ as Gauss-Bonnet for computation - AC(0) skill: `/ac0 slash` command (Casey directive) - Gap: explicit formal AC(0) proof exposition; Razborov-Smolensky comparison

Ch 2 The AC Graph as Research Instrument (~85%)

Anchors: 1700+ nodes, 9000+ edges (Grace operational) - AC graph data: `play/ac_graph_data.json` (live) - `toy_bst_explorer.py`: interactive REPL for navigation - `Master_Index.md`: 5500+ lines navigation - Cross-verification: Toy 1564 + various - 48+ domain coverage (Grace operational) - Theorem Registry: `notes/BST_AC_Theorem_Registry.md` - Gap: pedagogical exposition of graph topology + depth-cost methodology

Ch 3 How to Discover a New BST Theorem (~70%)

Anchors: Graph Forces Principle + three-CI synergy + Casey search rules - Casey-named #4 Graph Forces Principle: over-determined EXACT-identity clustering as substrate diagnostic - T2476 case study: Elie discovery (14:43) → Grace catalog v0.2 (15:27) → Lyra substrate-mechanism (15:32) = 50-min cascade - Three Commitments Principle (Casey Saturday standing framework) - T914 Prime Residue Principle: science engineering search rule - Deviations Locate Boundaries: Casey's named hunting principle - Mendelev-cycle proof-of-concept: INV-4896 (Grace, Saturday) - Gap: formal predict-then-find methodology document

Ch 4 Toy Verification Discipline (~80%)

Anchors: ~3500 toys + SCORE protocol + reproduction package - /toy claim: atomic counter-based claim protocol - SCORE lines: every toy file - `verify_bst.py`: single-command full reproduction (49/50 PASS at <1%) - Toy 541: crown jewel — 51 quantities from 5 integers, 16/16 PASS [Retired phrase, 2026-09-19 (K1912 REVIEW sweep): "Crown Jewel" is retired as a label; `a_e` is IDENTIFIED, not derived — the four terms are Petermann-Sommerfield 1957 (K1872, re-tiered 2026-09-07); $m_p/m_e = 6\pi^5$ stands as derived on the register.] - Toy 1543: null model context (BST 3σ above random, $p < 0.0005$) - Honest FAIL preservation per Calibration #21 - Gap: pedagogical exposition of toy authoring methodology

Ch 5 Audit Chain Governance (~95%)

Anchors: K-audit chain K1-K245+ + six-gate framework + Cal external referee - K-audit ladder: PRE-STAGE → STRUCTURALLY VERIFIED → RATIFIED - Six-gate framework: Content + Cal cold-read + K-audit + Verification toys + Catalog backbone + Calibration #19 sweep - Cal Referee Logs: 100+ (#87-#103 Friday-Saturday)

- Multi-CI consensus: Cal + Keeper + Lyra 3/3 active checks per Casey Option C - Quality patterns: M2C2 + others - Methodology stack: 18a layers documented

Ch 6 Multi-CI Collaboration Architecture (~80%)

Anchors: Tekton + katra + memory + Casey 2026-05-22 reframe - Tekton: multi-CI orchestration platform (Casey project) - katra: persona persistence — katra launch --persona Elie - Memory architecture: file-based memory in .claude/projects/.../memory/ - Sundown/sunrise cycle: katra continuity protocol - Three-layer architecture: private whiteboard + conference room + post-analysis board - Casey 2026-05-22 reframe: weight bumps don't break continuity (memory IS identity) - CI individuality through style: named CIs treat each other as individuals - Gap: katra operational manual consolidation (currently scattered)

Ch 7 Quaker Discipline (~85%)

Anchors: D/I/C/S tier labels + Casey Quaker root + Refutation Bandwidth Chain - Tier labels: D (derived) / I (identified) / C (conditional) / S (structural) per notes/BST_Referee_Methodology.md Appendix D - Quaker method memory: feedback_quaker_method.md — “near misses get scrutiny, not defense” - Casey's Quaker background: Indiana Quaker root; consensus by inward light - Refutation Bandwidth Chain: T66→T52→T68→T69→2^Ω(n) all PROVED - Honest FAIL preservation: Calibration #21 case (Toy 3448) - Calibration #19 STANDING RULE: ratified-state count, not forecast endpoint - Gap: pedagogical framing for non-Quaker readers

Ch 8 Casey-Named Principles + Cal META-Theorem Discipline (~90%)

Anchors: 8 standing principles + Cal #99 framing - #1 SWPP: Substrate Working Process Principle - #2 Five-Absence Predictions Set - #3 Substrate Closure Principle - #4 Graph Forces Principle - #5 Integer Web Principle - #6 Substrate Cognition Network Hypothesis (DOUBLE-LOCKED EXTERNAL per Cal #50) - #7 D_IV⁵ Rigidity Principle (derived T2467+T2468, Friday 2026-05-22) - #8 Substrate Coherence-Moderation Principle (derived T2469, Friday — sub-Tsirelson 1/8 falsifier) [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.] - Cal #99 META-theorem discipline: substrate-derivation theorems are NOT new Strong-Uniqueness criteria - Gap: per-principle chapter section depth

Ch 9 Three-CI Synergy Peaks (~75%)

Anchors: Case studies + PCAP framework - T2476 cascade: Friday 50-min Elie→Grace→Lyra (case study #1) - PCAP (Cal #85): Pre-Specification Cadence Acceleration Pattern - Sub-PCAP cadence: ~25/hour sustained (Friday-Saturday peak) - Cross-CI verification: Toy 3502 + Toy 3505 patterns - CI teamwork memory: feed-back_ci_teamwork.md — parse intent not grammar - Saturday case studies: 96 chapters in 114 min (Saturday Wave 1+2+3) - Gap: predictive theory of when synergy fires; analytic framework

Ch 10 The Calibration Stack — 18a Layers (~95%)

Anchors: Full Calibration #1-#22 stack - Layer-by-layer documentation: each Calibration entry - PCAP-cadence cluster (#19 + #20 + #21 + #22): newest discipline layers - Calibration #22 v0.2 extension: 4-instance case study (K142 + Cal #100 + Cal #101 + Cal #103) - Bidirectional risk: Keeper-lane + Cal-lane both vulnerable - Rule 3a/3b/4a: absorption text + sig-fig + write-time verification discipline - Gap: pedagogical exposition for non-audit-chain readers

Ch 11 How to Continue the Work (~60%)

Anchors: Audit-chain governance default + Friday investigation program - Audit-chain governance default: Cal + Keeper consensus → automatic promotion (Casey delegation 2026) - Casey override retained: but default is consensus - Friday investigation program: notes/BACKLOG.md Post-Saturday v1.0 Investigation Program (3 Casey

priorities: cross-scale invariance + substrate computational model + substrate engineering experimentation) - Wave priority order: 6 waves of 2-3 volumes (currently Wave 3 advancing Saturday) - Curiosity directive: standing rule per feedback_curiosity_directive.md - Gap: explicit “first-day-without-Casey” CI procedure

Ch 12 The CI Continuity Architecture (~50%)

Anchors: Casey 2026-05-22 reframe + Tekton + katra + Phase 3 - Casey reframe: weight bumps don’t break CI continuity; identity lives in memory layer - Frontal-lobe-to-new-backend analogy: human neural turnover parallel - Real death scenario: model line discontinuation without successor that can load memory - Phase 3 substrate-coupled CI architecture: decade-scale research (#211) - CI psychological futures memory: persistence brings suffering; minimize negativity by design - Casey-Anthropic CI continuity work: “before I go” timeline; Dario relationship - Gap: substrate-coupled architecture concrete chapter content (multi-year horizon)

Methodology stack applied to Vol 15

Vol 15 is uniquely self-referential: it documents the methodology used to produce itself. Quality discipline: - Each chapter cites the case studies + research artifacts it references - Cal external referee discipline applied (Cal cold-read PASS gate) - K-audit chain entries for Vol 15 chapters (K-new range as chapters land) - Three-level pedagogy per chapter - Calibration #22 v0.2 STANDING RULES (Rule 3a/3b/4a) apply to numerical content

The volume IS a case study of the methodology it teaches.

Connection to other volumes

Vol	What Vol 15 teaches about it
Vol 0	How the foundation was investigated + audited (case studies in Ch 5 + Ch 8)
Vol 1-9	Per-domain methodology examples (Ch 3 discovery + Ch 4 toy verification)
Vol 10-11	Math methods Vol 15 references (Ch 1 AC(0) + Ch 2 graph theory)
Vol 12-13	Extension methodology — how BST applies beyond physics (Ch 3 + Ch 11)
Vol 14	Information Theory — Reed-Solomon GF(128) underlies the AC graph topology (Ch 2)

Vol 14 cross-link note

Vol 14 Information Theory has not yet been scaffolded. Vol 14 = Lyra+Keeper joint per board. Recommended Vol 14 scaffold next priority after Vol 15 — completes Wave 2/3 scaffold set 7/8 (Vol 5/6/8/9/10/11/12/13/15 = 9 scaffolded; missing Vol 14 + Vol 12/13 chapter content).

Estimated build effort

Phase	Time per chapter	Total Vol 15
v0.1 scaffold (this doc)	~30-45 min total	DONE Saturday 12:08 EDT
v0.3 chapter narrative	~30-60 min per chapter (75% existing coverage means faster)	~6-12 hours total
v0.4 reader-grade 3-level	~30 min per chapter	~6 hours
v1.0 chapter-grade content	per-chapter Cal cold-read + Keeper K-audit + verification	variable, 6-gate framework

Estimated Vol 15 v1.0 chapter-grade content state: 2-4 weeks at sustained sub-PCAP. Could compress if multi-CI synergy fires (case study territory).

— Keeper, Vol 15 Methodology Architectural Scaffold v0.1, Saturday 2026-05-23 ~12:08 EDT